

Nicholas Nolen

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TECHNICAL SUMMARY

Master of Science in Computer Science student specializing in deep learning and neural data compression, graduating May 2027.

Languages: Java, C, C++, Rust, Python, SQL, Bash

Operating Systems: macOS, Windows, Linux

Frameworks: PyTorch, TensorFlow, Keras, React, Next.js, Spring Boot, Flask, SQLAlchemy

Cloud: GCP, Azure, Vercel

Software: IntelliJ, VSCode, Pycharm, Vim, Figma, Git, Docker

EDUCATION

Baylor University

Waco, TX

Master of Science in Computer Science

May 2027

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Multimedia Systems, Data Communications, Operating Systems, Advanced Database Design

Baylor University

Waco, TX

Bachelor of Science in Computer Science, GPA: 3.86 (Magna Cum Laude)

May 2026

- **Relevant Coursework:** Calculus II, Linear Algebra, Probability and Statistics, Discrete Structures, Systems Programming, Data Structures, Algorithms, Database Design, Cybersecurity, Technical Writing

PUBLICATIONS

UI-LIC: A Unified Framework for Evaluating Learned Image Compression Models

2026

- **Nicholas J. Nolen**, Luc Trudeau, Andrew C. Freeman. *ACM Multimedia (ACM MM) 2026* – **Accepted** (in final stage of picture-ready review).
- Engineered an open-source framework integrating six high-performance Learned Image Compression (LIC) models with a centralized controller for unified training, inference, and analysis alongside traditional video codecs.
- Developed a GUI and interactive image analyzer with configurable quality heatmap overlays to equalize bitrates and compute quality metrics (PSNR, SSIM, VMAF, LPIPS).

WORK EXPERIENCE

Modernization Developer

July 2026 – Present (20 hrs/week)

Baylor University

Waco, TX

- Upgraded web platform from **Python 2.7** to **Python 3.12** and **Django 4.2 LTS**, fixing broken syntax and updating database dependencies.
- Refactored legacy database models into modular packages to improve code organization and maintainability.
- Containerized the application with **Docker** and set up **GitLab CI/CD** pipelines for automated testing.

AI Product Engineering Intern

May 2025 – August 2025 (40 hrs/week)

Vanta ESports

Remote

- Designed AI coaching pipeline using **Rust** for telemetry object tracking, leveraged **Claude** for semantic gameplay analysis, and hosted on **Azure** cloud services; transitioned to **Web Assembly** for in-browser execution via **Vercel**.
- Presented solution to CEO and board, resulting in project being showcased for \$1.5M funding opportunity.

TECHNICAL PROJECTS

CNN Tuberculosis Diagnosis Predictor | *Python, Keras, TensorFlow, CNN* January 2026 – May 2026

- Engineered an automated preprocessing pipeline to extract 224×224 pixel crops centered on labeled regions of sputum smear slides across multiple inconsistent datasets.
- Architected a 5-layer CNN utilizing batch normalization and data augmentation to achieve a 0.9198 validation accuracy across varying slide qualities and staining conditions.
- Developed a "Crop Likelihood Viewer" to prioritize and visually aggregate high-probability viewfields, significantly reducing the manual search time for medical practitioners.

MarketOS (Farmers Market Capstone) | *Next.js, Spring Boot, SQL* January 2026 – May 2026

- Collaborated directly with client to define technical scope and requirements for a full-stack vendor management application, optimizing high-density transaction logging and addressing reporting inefficiencies.
- Developed a dynamic Excel template generation system and custom financial logic for vendor sales percentages, implementing an extensible labeling system to facilitate granular grant-tracking and auditing.

Bear Radio: Media Over QUIC Streaming Service | *Rust, QUIC, GCP, Opus* January 2025 – May 2025

- Engineered a terminal-based concurrent audio streaming service using **Media over QUIC (MoQ)** to enable low-latency, synchronized playback across multiple distributed devices.
- Developed publisher/subscriber architecture with **Opus** encoding and an interactive terminal UI, deploying the service via **GCP** to explore high-concurrency scaling and real-time data synchronization.

EXTRACURRICULAR ORGANIZATIONS

Baylor Chamber of Commerce August 2023 – December 2025

Secretary, Webmaster, Welcome Hour Coordinator, and Bear Committee Member

- Managed organization website using Drupal; supervised vendor relationships and budget for Welcome Hour.
- Cared for and helped feed the university's live black bear mascots within their on-campus habitat as a Bear Committee Member.